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Practical Data Modeling in Power BI: Beyond the Star Schema

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Geek by design

Because performance isn't optional

Four Problems Beyond the Star Schema

- Grain — the decision that shapes everything
- Fact type — transaction, snapshot, accumulating
- History — Slowly Changing Dimensions (SCD2)
- Many-to-many — bridge, or an alternative

Our Roadmap

Four problems, one continuous model — each with a price.

- Grain — the decision that shapes everything
- Fact type — transaction, snapshot, accumulating
- History — Slowly Changing Dimensions (SCD2)
- Many-to-many — bridge, or an alternative?

Granularity Determines Your Model

Ask three questions

- What business question?
- What event are we measuring?
- What does one row represent?

Grain decides everything downstream

- The dimensions
- The valid measures
- The fact type

Unclear grain → unstable model.

What Goes Wrong When Granularity Is Wrong?

When grains don't align

- Different detail levels compared directly
- Totals look correct — but are conceptually wrong

Example: monthly target vs daily sales

- Two grains: target = month, sales = day
- Repeating the target per day inflates the total

This is exactly Demo 1 — up next.

Demo Time

Grain mismatch, live. No safety net.

Granularity Drives Fact Type

Transaction Fact

- One row per business event
- Detailed, event-driven data

Snapshot Fact

- One row per entity per time period
- Captures status at a specific moment

Accumulating Snapshot Fact

- One row per process instance
- Tracks milestones within a lifecycle

Transaction Fact

Definition

- One row per business event

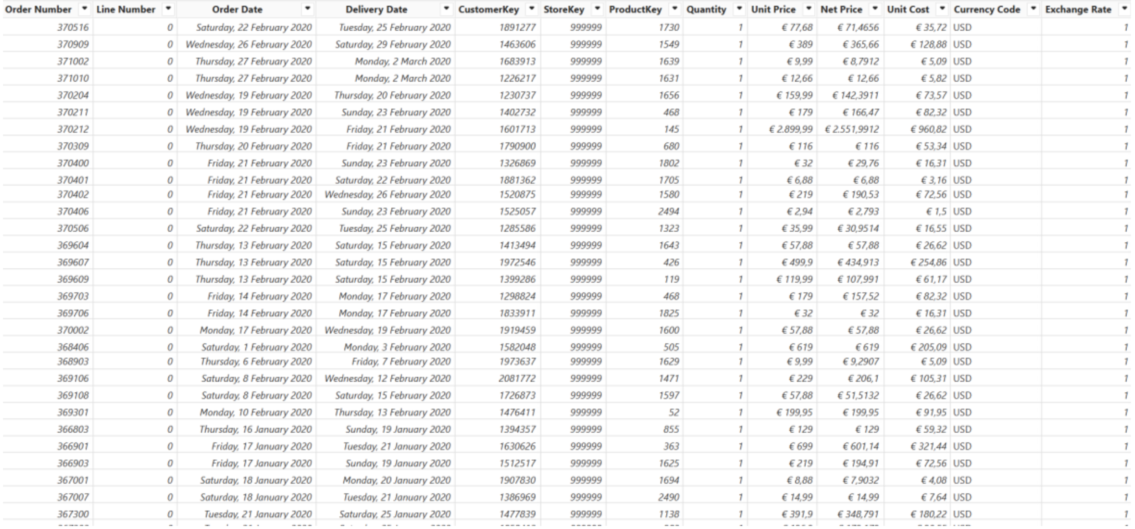
Grain

- Atomic level of detail
- Clearly defined business event
- No mixed levels of detail

Structure

- Foreign keys to dimensions
- Numeric, usually fully additive measures
- Large number of rows

A transaction fact stores the lowest meaningful level of detail.



Order Number	Line Number	Order Date	Delivery Date	CustomerKey	StoreKey	ProductKey	Quantity	Unit Price	Net Price	Unit Cost	Currency Code	Exchange Rate
370516	0	Saturday, 22 February 2020	Tuesday, 25 February 2020	1891277	999999	1730	1	€ 77,68	€ 71,4656	€ 35,72	USD	1
370909	0	Wednesday, 26 February 2020	Saturday, 29 February 2020	1463606	999999	1549	1	€ 389	€ 365,66	€ 128,88	USD	1
371002	0	Thursday, 27 February 2020	Monday, 2 March 2020	1683913	999999	1639	1	€ 9,99	€ 8,7912	€ 5,09	USD	1
371010	0	Thursday, 27 February 2020	Monday, 2 March 2020	1226217	999999	1631	1	€ 12,66	€ 12,66	€ 5,82	USD	1
370204	0	Wednesday, 19 February 2020	Thursday, 20 February 2020	1230737	999999	1656	1	€ 159,99	€ 142,3911	€ 73,57	USD	1
370211	0	Wednesday, 19 February 2020	Sunday, 23 February 2020	1402732	999999	468	1	€ 179	€ 166,47	€ 82,32	USD	1
370212	0	Wednesday, 19 February 2020	Friday, 21 February 2020	1601713	999999	145	1	€ 2.899,99	€ 2.551,9912	€ 960,82	USD	1
370309	0	Thursday, 20 February 2020	Friday, 21 February 2020	1790900	999999	680	1	€ 116	€ 116	€ 53,34	USD	1
370400	0	Friday, 21 February 2020	Sunday, 23 February 2020	1326869	999999	1802	1	€ 32	€ 29,76	€ 16,31	USD	1
370401	0	Friday, 21 February 2020	Saturday, 22 February 2020	1881362	999999	1705	1	€ 6,88	€ 6,88	€ 3,16	USD	1
370402	0	Friday, 21 February 2020	Wednesday, 26 February 2020	1520875	999999	1580	1	€ 219	€ 190,53	€ 72,56	USD	1
370406	0	Friday, 21 February 2020	Sunday, 23 February 2020	1525057	999999	2494	1	€ 2,94	€ 2,793	€ 1,5	USD	1
370506	0	Saturday, 22 February 2020	Tuesday, 25 February 2020	1285586	999999	1323	1	€ 35,99	€ 30,9514	€ 16,55	USD	1
369604	0	Thursday, 13 February 2020	Saturday, 15 February 2020	1413494	999999	1643	1	€ 57,88	€ 57,88	€ 26,62	USD	1
369607	0	Thursday, 13 February 2020	Saturday, 15 February 2020	1972546	999999	426	1	€ 499,9	€ 434,913	€ 254,86	USD	1
369609	0	Thursday, 13 February 2020	Saturday, 15 February 2020	1399286	999999	119	1	€ 119,99	€ 107,991	€ 61,17	USD	1
369703	0	Friday, 14 February 2020	Monday, 17 February 2020	1298824	999999	468	1	€ 179	€ 157,52	€ 82,32	USD	1
369706	0	Friday, 14 February 2020	Monday, 17 February 2020	1833911	999999	1825	1	€ 32	€ 32	€ 16,31	USD	1
370002	0	Monday, 17 February 2020	Wednesday, 19 February 2020	1919459	999999	1600	1	€ 57,88	€ 57,88	€ 26,62	USD	1
368406	0	Saturday, 1 February 2020	Monday, 3 February 2020	1582048	999999	505	1	€ 619	€ 619	€ 205,09	USD	1
368903	0	Thursday, 6 February 2020	Friday, 7 February 2020	1973637	999999	1629	1	€ 9,99	€ 9,2907	€ 5,09	USD	1
369106	0	Saturday, 8 February 2020	Wednesday, 12 February 2020	2081772	999999	1471	1	€ 229	€ 206,1	€ 105,31	USD	1
369108	0	Saturday, 8 February 2020	Saturday, 15 February 2020	1726873	999999	1597	1	€ 57,88	€ 51,5132	€ 26,62	USD	1
369301	0	Monday, 10 February 2020	Thursday, 13 February 2020	1476411	999999	52	1	€ 199,95	€ 199,95	€ 91,95	USD	1
366803	0	Thursday, 16 January 2020	Sunday, 19 January 2020	1394357	999999	855	1	€ 129	€ 129	€ 59,32	USD	1
366901	0	Friday, 17 January 2020	Tuesday, 21 January 2020	1630626	999999	363	1	€ 699	€ 601,14	€ 321,44	USD	1
366903	0	Friday, 17 January 2020	Sunday, 19 January 2020	1512517	999999	1625	1	€ 219	€ 194,91	€ 72,56	USD	1
367001	0	Saturday, 18 January 2020	Monday, 20 January 2020	1907830	999999	1694	1	€ 8,88	€ 7,9032	€ 4,08	USD	1
367007	0	Saturday, 18 January 2020	Tuesday, 21 January 2020	1386969	999999	2490	1	€ 14,99	€ 14,99	€ 7,64	USD	1
367300	0	Tuesday, 21 January 2020	Saturday, 25 January 2020	1477839	999999	1138	1	€ 391,9	€ 348,791	€ 180,22	USD	1

Snapshot Fact

Definition

- One row per entity per time period

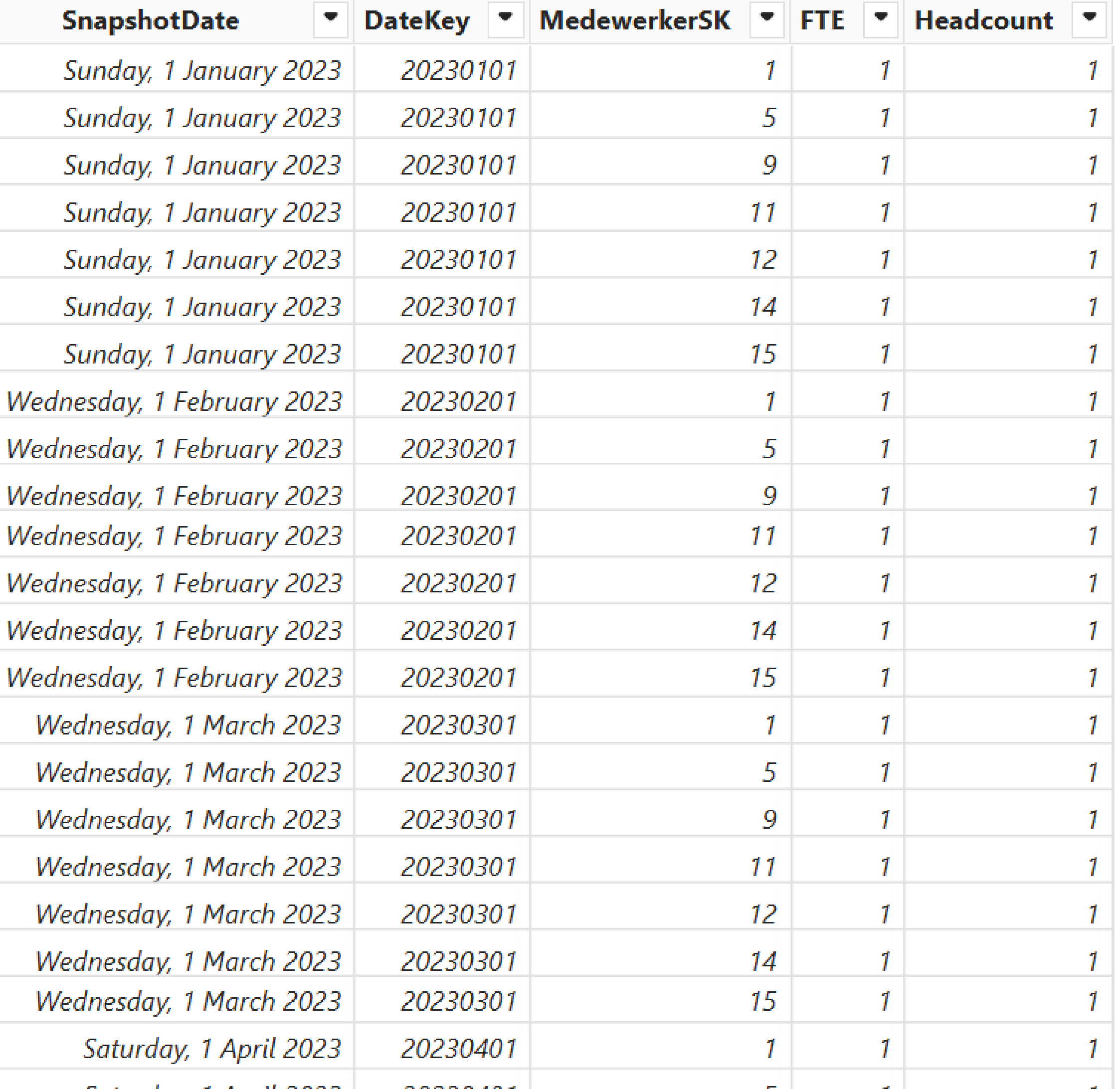
Grain

- Status at a specific moment in time
- One entity per defined period (e.g., day, month)
- Time-based consistency

Structure

- Foreign keys to dimensions
- Measures represent a state, not an event
- Often semi-additive

A snapshot fact captures a state — not an event.



SnapshotDate	DateKey	MedewerkerSK	FTE	Headcount
Sunday, 1 January 2023	*20230101*	*1*	*1*	*1*
Sunday, 1 January 2023	*20230101*	*5*	*1*	*1*
Sunday, 1 January 2023	*20230101*	*9*	*1*	*1*
Sunday, 1 January 2023	*20230101*	*11*	*1*	*1*
Sunday, 1 January 2023	*20230101*	*12*	*1*	*1*
Sunday, 1 January 2023	*20230101*	*14*	*1*	*1*
Sunday, 1 January 2023	*20230101*	*15*	*1*	*1*
Wednesday, 1 February 2023	*20230201*	*1*	*1*	*1*
Wednesday, 1 February 2023	*20230201*	*5*	*1*	*1*
Wednesday, 1 February 2023	*20230201*	*9*	*1*	*1*
Wednesday, 1 February 2023	*20230201*	*11*	*1*	*1*
Wednesday, 1 February 2023	*20230201*	*12*	*1*	*1*
Wednesday, 1 February 2023	*20230201*	*14*	*1*	*1*
Wednesday, 1 February 2023	*20230201*	*15*	*1*	*1*
Wednesday, 1 March 2023	*20230301*	*1*	*1*	*1*
Wednesday, 1 March 2023	*20230301*	*5*	*1*	*1*
Wednesday, 1 March 2023	*20230301*	*9*	*1*	*1*
Wednesday, 1 March 2023	*20230301*	*11*	*1*	*1*
Wednesday, 1 March 2023	*20230301*	*12*	*1*	*1*
Wednesday, 1 March 2023	*20230301*	*14*	*1*	*1*
Wednesday, 1 March 2023	*20230301*	*15*	*1*	*1*
Saturday, 1 April 2023	*20230401*	*1*	*1*	*1*

Accumulating Snapshot Fact

Definition

- One row per process instance

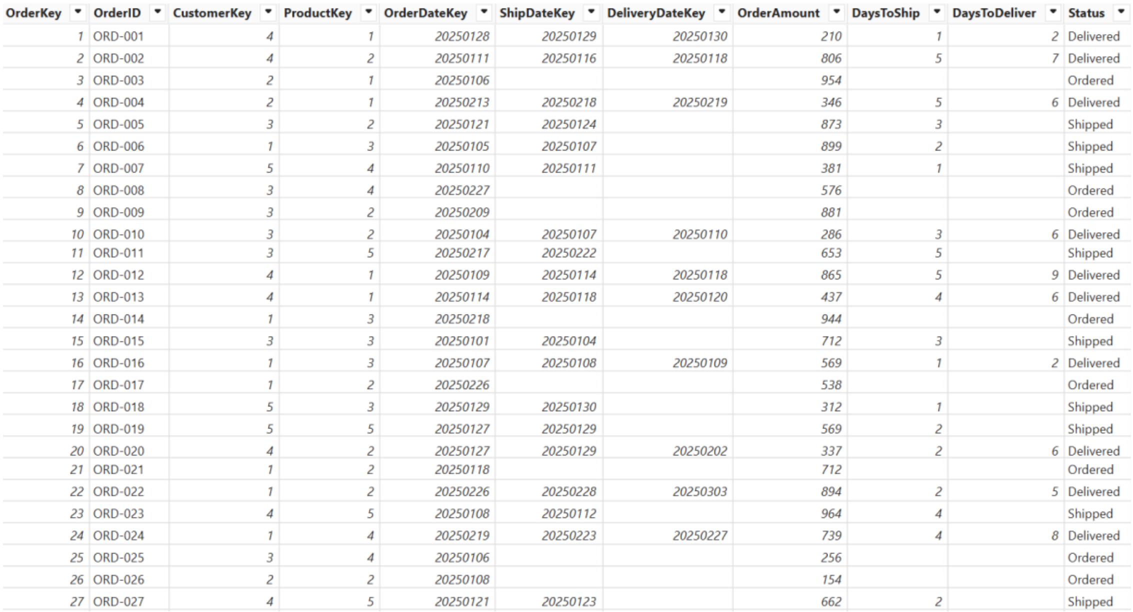
Grain

- One business process
- Multiple milestones tracked in a single row
- Updated as the process progresses

Structure

- Foreign keys to dimensions
- Multiple date columns (milestones)
- Measures often represent durations or cycle times

An accumulating snapshot tracks progress — not repeated events.



OrderKey	OrderID	CustomerKey	ProductKey	OrderDateKey	ShipDateKey	DeliveryDateKey	OrderAmount	DaysToShip	DaysToDeliver	Status
1	ORD-001	4	1	20250128	20250129	20250130	210	1	2	Delivered
2	ORD-002	4	2	20250111	20250116	20250118	806	5	7	Delivered
3	ORD-003	2	1	20250106			954			Ordered
4	ORD-004	2	1	20250213	20250218	20250219	346	5	6	Delivered
5	ORD-005	3	2	20250121	20250124		873	3		Shipped
6	ORD-006	1	3	20250105	20250107		899	2		Shipped
7	ORD-007	5	4	20250110	20250111		381	1		Shipped
8	ORD-008	3	4	20250227			576			Ordered
9	ORD-009	3	2	20250209			881			Ordered
10	ORD-010	3	2	20250104	20250107	20250110	286	3	6	Delivered
11	ORD-011	3	5	20250217	20250222		653	5		Shipped
12	ORD-012	4	1	20250109	20250114	20250118	865	5	9	Delivered
13	ORD-013	4	1	20250114	20250118	20250120	437	4	6	Delivered
14	ORD-014	1	3	20250218			944			Ordered
15	ORD-015	3	3	20250101	20250104		712	3		Shipped
16	ORD-016	1	3	20250107	20250108	20250109	569	1	2	Delivered
17	ORD-017	1	2	20250226			538			Ordered
18	ORD-018	5	3	20250129	20250130		312	1		Shipped
19	ORD-019	5	5	20250127	20250129		569	2		Shipped
20	ORD-020	4	2	20250127	20250129	20250202	337	2	6	Delivered
21	ORD-021	1	2	20250118			712			Ordered
22	ORD-022	1	2	20250226	20250228	20250303	894	2	5	Delivered
23	ORD-023	4	5	20250108	20250112		964	4		Shipped
24	ORD-024	1	4	20250219	20250223	20250227	739	4	8	Delivered
25	ORD-025	3	4	20250106			256			Ordered
26	ORD-026	2	2	20250108			154			Ordered
27	ORD-027	4	5	20250121	20250123		662	2		Shipped

Demo Time

Snapshot facts: periodic & accumulating, live.

When Granularity and Fact Types Aren’t Enough

Even with the correct grain and fact type…

- The model is right — and results can still be wrong

Why? Dimensions change over time

- History is lost
- Facts get evaluated against today's dimension values

This is what SCD2 solves — up next.

What Are Slowly Changing Dimensions (SCD)?

A dimension that changes over time

- Employee changes department · product changes category · customer moves region

The core question

Overwrite history — or preserve it?

The Three Types of SCD

Type 1 — Overwrite

- No history — only the current state

Type 2 — Preserve history

- New row per change; facts link to the right version

Type 3 — Limited history

- Previous value kept in an extra column

Enter SCD Type 2

How it works

- New row per change, each with its own SurrogateKey
- Facts link to the correct historical version

Typical columns

- SurrogateKey · BusinessKey · ValidFrom / ValidTo · IsCurrent

Why it matters

- “Employee was in Department A at the time of the sale”

Demo Time

SCD Type 2: preserve history — and the trap, live.

What Are Many-to-Many Relationships (M2M)?

A many-to-many relationship

- One value in A → many in B, and one in B → many in A — no unique match

Why it breaks the star schema

- Facts are normally many-to-one to dimensions
- M2M brings: ambiguous filters, double counting, complex DAX, perf cost

Why M2M Breaks Your Model

Filter propagation is hard to control

- No longer strictly many-to-one — result depends on context

Rows don’t add up to the total

- One fact → many dimension rows; every row is right, their sum isn’t

Grain mismatch is often the root cause

- Budget vs sales — the relationship compensates for a modeling issue

Demo Time

Many-to-many via a bridge, live.

Takeaways

Every modeling choice has a price. Choose it on purpose.

- Grain drives the fact type; the fact type drives your DAX
- Every choice has an engine price: speed, memory, or correctness
- The price moves, it never disappears — choose on purpose
- Speed ≠ correctness; the engine won't warn you

Questions? Let’s connect

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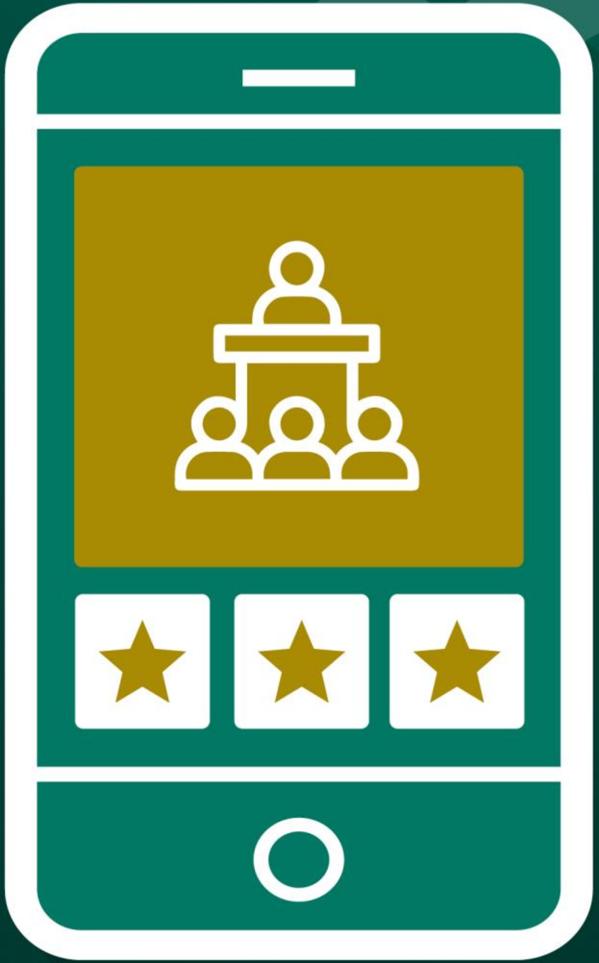

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